

Cooling Water System — Pump P-201A

Operations & Reliability Manual

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1. Scope & System Overview

This manual governs the operation, condition monitoring, and reliability maintenance of Centrifugal Pump **P-201A**, the primary circulation pump for the Cooling Water System. P-201A draws suction through Gate Valve **V-104** and discharges to the main cooling header.

2. Equipment Description

P-201A is a horizontal, single-stage centrifugal pump rated for 450 m³/hr at 55 m total dynamic head, driven by a 132 kW induction motor at 2,950 RPM. The unit is fitted with a single mechanical seal and grease-lubricated antifriction bearings.

3. Condition Monitoring

- Monthly vibration readings on P-201A shall not exceed 4.5 mm/s RMS per ISO 10816-3; any reading above this threshold requires an immediate work order for further inspection before continued operation.
- Bearing temperatures shall be logged weekly and shall not exceed 80°C at the drive-end housing under normal load.

4. Maintenance & Reliability

- Replace the mechanical seal cartridge on P-201A every 18 months, or immediately upon any visible leakage at the seal housing, whichever occurs first.
- After any seal or bearing replacement, verify shaft alignment between P-201A and its driver motor is within 0.05 mm TIR using a dial indicator or laser alignment method before returning the unit to service.

5. Compliance References

P-201A is a centrifugal pump selected, installed, and maintained in accordance with API Standard 610 for petroleum, heavy-duty chemical, and gas industry services. Mechanical integrity activities on P-201A are tracked under the facility's OSHA 1910.119 Process Safety Management program.